10 Binomial Distributions

- Some probabilistic situations are so common that they have special names.
- In a sequence of **Bernoulli(p) trials**
  - There are only two possible outcomes, "success" (1) or not (0, "failure"), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to p.
  - The trials are independent.
- There are several commonly used distributions associated with Bernoulli trials, including Geometric and Binomial.

Example 10.1. Consider an extremely simplified model for the daily closing price of a certain stock. Every day the price either goes up or goes down, and the movements are independent from day-to-day. Assume that the probability that the stock price goes up on any single day is 0.55. Let X be the number of days on which the price goes up in a 5-day week.

1. Compute and interpret $\mathrm{P}(X = 0)$.

2. Compute the probability that the price goes up on the first day and then down on the following four days.

3. Why is $\mathrm{P}(X = 1)$ different from the probability in the previous part? Compute and interpret $\mathrm{P}(X = 1)$.

4. Compute and interpret $\mathrm{P}(X = 2)$.

5. Suggest a general formula for the probability mass function of X.

6. Construct a table and plot to represent the distribution of X, and interpret it.

- Consider a *fixed number*, n, of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
- A Binomial distribution is specified by two parameters:
  - n (an integer): the fixed number of Bernoulli trials, or the sample size
  - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Binomial distribution** with parameters n, a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, \dots, n$$

  - In the Binomial(n, p) situation, there are $\binom{n}{x}$ S/F sequences that result in exactly x successes—and thus $n - x$ failures—in n trials,
  - and each of these sequences has probability $p^x(1-p)^{n-x}$
- The *binomial coefficient* (read "n choose x")

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

counts the number of success/failure sequences of length n in which there are exactly x successes. (Remember: by definition $0! = 1$.)

Example 10.2. Continuing Example 10.1.

1. Suggest a shortcut formula for $\mathrm{E}(X)$.

2. Compute $\mathrm{E}(X)$ using the distribution of X from Example 10.1. Did the shortcut formula work?

3. Interpret $\mathrm{E}(X)$.

4. Remember that we assumed $p = 0.55$. If instead p were 0.9, would $\mathrm{Var}(X)$ be greater or less? What if $p = 0.5$?

- If X has a Binomial(n, p) distribution then

$$\begin{aligned}\mathrm{E}(X) &= np\\ \mathrm{Var}(X) &= np(1-p)\end{aligned}$$

10.1 Exercises

Exercise 10.1. Continuing Example 10.1.

1. What does the random variable $5 - X$ represent? What is its distribution?

2. Suppose that the price is currently \$100 and each it either moves up \$2 or down \$2. Let S be the stock price after 5 days. How does S relate to X? Does S have a Binomial distribution?

3. Recall that X is the number of days on which the price goes up in the next five days. Suppose that Y is the number of days on which the price goes up in the ten days after that (days 6-15). What is the distribution of Y? What does $X + Y$ represent and what is its distribution? (Continue to assume independence between days, with probability 0.55 of an up movement on any day.)

- If X and Y are independent, X has a Binomial(n_X, p) distribution, and Y has a Binomial(n_Y, p) distribution then $X + Y$ has a Binomial($n_X + n_Y$, p)
- A Binomial(1, p) distribution is also known as a **Bernoulli(p)** distribution, taking a value of 1 with probability p and 0 with probability $1 - p$.
  - For a single trial, a Bernoulli random variable *indicates* if there is a success (1) or failure (0) on the trial
- If $X_1, X_2, \ldots, X_n$ are independent each with a Bernoulli(p) distribution, then $X_1 + \cdots + X_n$ has a Binomial(n, p) distribution.
  - The random variable $X_1 + X_2 + \ldots + X_n$ incrementally counts the total number of successes in the n trials

Exercise 10.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p. If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.

2. Roll a die 20 times; X is the number of times the die lands on 6.

3. Roll a die until it lands on 6; X is the total number of rolls.

4. Roll a die 20 times; X is the sum of the numbers rolled.

5. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)

6. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).

7. Randomly select a sample of 35 Cal Poly students; X is the number of students in the sample who are CA residents.

- Consider selecting a random sample of size n from a population in which every individual is classified as "success" or "failure". Suppose that the proportion of individuals in the population classified as success is p.
- The individuals selected for the sample can be considered a sequence of Bernoulli(p) trials.
  - There are only two possible outcomes, success (1) and failure (0), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to the population proportion of success p
  - The trials are independent
    * If sampling with replacement, or
    * If sampling without replacement but the population size is much larger than the sample size n (which is usually the case in practice)
- If X counts the number of successes in a random sample of size n from a large population with population proportion of success p then X has a Binomial(n, p) distribution.

Exercise 10.3. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution*. Can you think of X and Y that have the same distribution but for which $\mathrm{P}(X = Y) = 0$?